

Module 12 – DevOps and CI/CD Overview

Introduction to DevOps

What is DevOps?

DevOps is a combination of **Development (Dev)** and **Operations (Ops)**. It is a software development approach that promotes collaboration between development and operations teams to build, test, and deploy applications faster and more efficiently. DevOps uses automation and continuous processes to improve software quality and speed up software delivery.

Goals of DevOps

- Improve collaboration between development and operations teams.
- Deliver software faster.
- Automate development and deployment tasks.
- Improve software quality.
- Reduce deployment failures.
- Enable faster bug fixes.
- Increase customer satisfaction.

Benefits of DevOps

- Faster software delivery.
- Better collaboration between teams.
- Improved code quality.
- Reduced manual work through automation.
- Faster issue detection and recovery.
- Reliable deployments.
- Increased productivity and scalability.

Key DevOps Practices

- **Continuous Integration (CI):** Frequently merge code into a shared repository where it is automatically built and tested.
- **Continuous Delivery (CD):** Automatically prepares software for deployment after successful testing.
- **Continuous Deployment:** Automatically deploys every successful code change to production.
- **Continuous Testing:** Automated testing throughout development.
- **Continuous Monitoring:** Monitors application performance and detects issues.

- **Infrastructure as Code (IaC):** Manages infrastructure using code instead of manual configuration.
 - **Collaboration and Communication:** Encourages teamwork across development and operations.
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Understanding CI/CD

What is Continuous Integration (CI)?

Continuous Integration (CI) is the practice of automatically building and testing code whenever developers commit changes to a shared repository. It helps detect bugs early and ensures that code changes integrate smoothly.

Benefits of CI

- Early bug detection.
- Improved code quality.
- Faster development.
- Better collaboration.
- Reduced integration issues.

What is Continuous Delivery (CD)?

Continuous Delivery (CD) automatically prepares software for deployment after successful testing. The application is always ready for release, but deployment to production usually requires manual approval.

Benefits of Continuous Delivery

- Faster releases.
- Reliable deployments.
- Lower deployment risk.
- Better software quality.

What is Continuous Deployment?

Continuous Deployment automatically deploys every successful code change directly to production without manual approval.

Benefits of Continuous Deployment

- Faster software updates.
- Immediate customer feedback.
- Fully automated deployments.

Difference Between CI and CD

Continuous Integration (CI)	Continuous Delivery (CD)
Integrates code frequently.	Prepares software for deployment.
Automatically builds and tests code.	Builds, tests, and packages applications.
Detects bugs early.	Ensures software is always deployment-ready.

Benefits of CI/CD

- Faster software delivery.
 - Automated testing and deployment.
 - Better code quality.
 - Reduced manual effort.
 - Reliable releases.
 - Faster recovery from failures.
 - Improved customer satisfaction.
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CI/CD Tools and Platforms

1. Jenkins

Jenkins is an open-source automation server used for Continuous Integration and Continuous Delivery.

Features

- Open-source.
- Supports build, testing, and deployment.
- Large plugin ecosystem.
- Highly customizable.

2. GitHub Actions

GitHub Actions is GitHub's built-in CI/CD platform that automates workflows directly from GitHub repositories.

Features

- Integrated with GitHub.
- Uses YAML workflow files.
- Automates testing and deployment.
- Easy to configure.

3. GitLab CI/CD

GitLab CI/CD is a built-in DevOps platform that supports automated testing, deployment, and security scanning.

Features

- Built-in CI/CD pipelines.
- Deployment automation.
- Security scanning.
- Container and Kubernetes support.

4. CircleCI

CircleCI is a cloud-based CI/CD platform that automates software development workflows.

Features

- Fast cloud builds.
- Parallel testing.
- Docker and Kubernetes support.
- YAML-based configuration.

Comparison of Popular CI/CD Tools

Tool	Purpose
Jenkins	Open-source CI/CD automation
GitHub Actions	CI/CD for GitHub repositories
GitLab CI/CD	Complete DevOps platform
CircleCI	Cloud-based CI/CD automation

Summary

DevOps is a modern software development approach that combines development and operations to improve collaboration, automate workflows, and deliver software faster. Continuous Integration (CI) focuses on automatically building and testing code, while Continuous Delivery (CD) prepares software for deployment and Continuous Deployment automates deployment to production. Popular tools such as **Jenkins**, **GitHub Actions**, **GitLab CI/CD**, and **CircleCI** help organizations automate software development, improve code quality, and deliver reliable applications more efficiently.